



Tours and Travel Agency Web Based System For Live Lanka Tours

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Abstract

Live Lanka Tours is a tourism company with some unique tour packages which is currently located in Polonnaruwa Dalukana area as their head office.

Currently they are doing their business through trip advisor account and Facebook page and they communicate with their customers by using WhatsApp and Messenger apps. They provide their service only for foreign customers/tourists. Therefore, current process is deficiency for their business. They do not maintain proper paper work and also it is very hard to keep customer details and other reports.

Because of these problems, the owner has decided to use a web based system. This web based system will support to manage and monitor their work process properly. Tourists can select tour from the given tour packages or they can request new tours through this proposed system. And also the system will provide various facilities for company and their customers. Such as user login, user profile with their booking history, booking tours, make online payments, generate reports and administrator functions. This system will give a customer satisfaction and efficient way in doing their work.

Accommodation owners, vehicle owners, drivers and guides also can register to the company through the system. But they must have qualified the company requirements. After registration process they can log in to the system and they can view their profile with their tour/booking history. And also system will update their new tours and booking notifications. They can accept or decline those bookings and tours.

PHP, JavaScript, AJAX, jQuery, CSS are using as the programming languages to develop this web based system. Bootstrap is the selected framework and Apache NetBeans is selected as the IDE. MySQL and Apache web servers are using as the database designing tools.

List of acronyms

PHP - PHP: Hypertext Preprocessor

SQL - Structured Query Language

AJAX - Asynchronous JavaScript and XML

CSS - Cascading Style Sheet

IT - Information Technology

URL - Uniform Resource Locator

DFD - Data Flow Diagram

UML - Unified Modelling Language

HTML - Hypertext Markup Language

HTTP - Hypertext Transfer Protocol

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Chapter 01 - Introduction

1.1 Introduction to Live Lanka Tours

Sri Lanka has a growing tourism industry and it has been popular place of attraction for foreign travelers, especially because of the natural beauty of the geographical locations of the country and also Sri Lankan's hospitality and Sri Lankan's foods.

Live Lanka Tours is a travel company who provides their services to the foreign tourists. Earlier they named their company as Elka Tours Travel Agency. But while their registration process due to some reason they changed company name as Live Lanka Tours. The company is currently located in Polonnaruwa Dalukana area in Sri Lanka.

1.2 Motivation for the project

Globalization has made tourism a popular global leisure activity. Sri Lanka has a growing tourism industry and it has been popular place of attraction for foreign travelers, especially because of the natural beauty of the geographical locations of the country and also Sri Lankan's hospitality and Sri Lankan's foods. In 2019 Sri Lanka has been named the best country in the world to visit by Lonely Planet.

Live Lanka Tours is a new growing travel company who provides their services to the foreign tourists. But still the company does not have a web based system and currently they are doing their business through Facebook page and also current process is deficiency for their business.

They do not maintain proper paperwork also. Therefore, the company currently facing tour booking, tour package management and customer management issues, data security issues and data redundancy issues. Not only that but also; current process takes more time to respond for questions asked by customers and also it takes more time to accept tour bookings. In the current process the customer cannot arrange customized tours. Only they can book already provided ready-made tour packages from the company. Therefore, this current process is very boredom way to the customer and the company. Because of those reason the owner interested in a web based system.

To overcome above issues "Live Lanka Tours" web based system will be a perfect solution for them. This system will help them to work effectively, maintain proper customer management and provide their best service to the customer.

Successfully completing the final individual project is the key component of the Bachelor of Information Technology degree program of University of Colombo School of Computing, and it is a golden opportunity to apply practical knowledge that I have gained throughout the degree, and also a pass with a good grade in the final project will be an added advantage to lead a successful career path in future.

And also it's an opportunity to prove myself as a profession that I gain through the course and also a chance me to interact with the real clients in it environments.

1.3 Objectives of the system

Tours and travel agency web based system is a web based application and maintains a centralized repository of all the related information. The objective of the project is to develop a system that automates the process and activities of the travel agency. The

purpose is to design a system using which one can perform all operations related to traveling and sightseeing.

- The system provides facilities to connect directly customers and agent through internet.
- The system provides registration facilities to tourist, accommodation owners, guides and vehicle owners.
- The system provides facilities to create, modify and delete ready-made tour packages.
- The system provides facilities to find various ready-made tour packages all over the Sri Lanka.
- The system provides facilities to registered tourists to book ready-made tour packages.
- The system provides facilities to registered tourists to request customized tour packages.
- The system provides facilities to registered tourists to track their booking process.
- This system provides facilities to find accommodations details, transportation details and guide details which are already registered with the company The system displays attractive tourist places.
- This system provides facilities for manager to maintain their quality management sessions and after done this session accommodation owners, vehicle owners and guides can confirm and/or continue their registration with the company.
- The system provides facilities to accommodation owners, vehicle owners and guides can update availability and also view their upcoming tour bookings.
- The system provides facilities for all system users to manage their profile and view tour history.
- The system provides facilities for manager to calculate accommodation, transportation and guide charges.
- This system provides feedback mechanism for tourist.
- This system provides company official social media links.
- This system will help to makes strong relationship with customers so that they can enjoy that holiday of their dreams.
- This system also helps to develop tourism with different cultures so that they enrich the tourists experience and build pride.

Finally, I can say this system will provide easy and efficient work for the company.

1.4 Scope of the system

Tours and travel management system is an integrated software developed for the Live Lanka Tours Company. The main aim of this project is to help the Live Lanka Tours Company to manage their customers, tour packages, transportation, guides and accommodation. It makes all operation of the tour company easy and accurate. The web based system makes tourism management easy by handling customers' requests and providing servers for the customers located at different parts of the various cities. Different modules have been incorporated in this project to handle different parts and sector of the tour management field.

- **Registration module**
Tourists, accommodation owners, vehicle owners and guides all these users have different registration process. After registration they can log-in into the system. They have different dashboards. All those users can manage their profiles and view their booking history.
- **Booking module**
Company provides two types of tour package booking processes. Those are ready-made tour bookings and customized tour bookings. The company provides various types of ready-made tour packages. Registered tourists can book ready-made tour package if they want. Otherwise they can request customized tour package.
- **Payment module.**
Customer can do the payment while he/she booking a ready-made tour package. Customer must upload bank details and a bank payment slip or an online bank transfer receipt. If not ready-made tour booking process cannot complete. In customized tour booking, when the company confirmed their tour request customer can do the payment and they must upload the payment details and slip or receipt within 24 hours.
- **View notification module**
Customer can view accepted email, finalized tour receipt and also declined emails with declined reasons.
- **Tracking module**
When a customer booked a tour or requested a tour customer can track their booking process. After booking a ready-made tour package customer can track the booking process whether it is pending, viewed or accepted. If a customer requests a customized tour package, customer can track the booking process whether it is pending, viewed, request accepted, pending payment or confirm payment details and accepted tour.
- **Quality management session module**
In this module accommodation owner, vehicle owner, and guide will interview by the company and also vehicle check-ups and accommodation visit are doing by the company. If they pass all those processes they will confirmed as a registered guide, accommodation or vehicle with the company. Once in a six month this process will doing by the company. Days will update automatically by the system. And also if they did not cooperate with the company terms and regulation the company will terminate and backlist them.
- **Manage availability module**
Accommodation owners, vehicle owners and guides can update their availability. Manager/admin can view their availability and also customer only view available accommodations, transportations and guides while their booking process.
- **Calculate commission and payment module**
Manager can view accommodation charges, transportation charges and guide charges. System will calculate those charges automatically and notify manager.
- **Helpdesk module**
Customers can find FAQs in here and also willing to add live chat facility for customer.

- Report management and decision making module
System generates certain set of reports (Automated). Such as monthly booked ready-made tour packages, monthly booked customized tour packages, annual registered guides, accommodations, and vehicle, monthly guide, accommodation and vehicle charges and etc.
- Reviews and feedbacks module
Customer can comment their ideas and suggestions. Then customers can review customer feedback through the comments.
- Search and find module
Customers can search and find details of guides, accommodations and transportations which are registered with the company.

1.5 Structure of the dissertation

The dissertation covers the detailed description of the Live Lanka Tours and travel agency web based system and the rest of chapter's structure as follows.

Chapter 01 -Introduction

In this chapter I briefly explain the introduction about the proposed system. Current process and the background. Motivation for the project. Objectives and scope for the proposed system.

Chapter 02-Analysis

In this chapter I describe requirement gathering method and analyzing techniques of the project. I mentioned existing similar systems which were I referring to develop this proposed system and what are the functional and non-functional requirements of the system.

Chapter 03-Design

In this chapter I gave few user-interface design and system design with diagrams.

Chapter 04-Implementation

In this chapter I gave an explanation about what were the using hardware, software, development tools and the techniques of the proposed system.

Chapter 05-Evaluation

In this chapter I explained what were the different testing techniques and test cases to specify the system testing also with user evaluations.

Chapter 06-Conclusion

In this chapter I explained the success of the project. Such as which functionality support with which client requirement. Future modifications also including in this chapter

Chapter 02 - Analysis

2.1 Introduction

System analysis is one of the main phase in the software development life cycle. This phase gives an overall image of the system and also this phase give a high level description of the system. Main objectives of this phase are what services system should provide, required performance of the system.

Before analyzing the system, requirements should be gathered by using the fact finding techniques, such as interviews, observations, sample documents etc. And also system functional requirements and non-functional requirements are discussing in this phase.

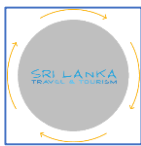
2.2 Reviews of similar system with references

The investigation of comparative web based systems helped to understand the good interface properties, the essential highlights of the systems and how the required functionalities are presented. More experience could also be gained by the practice of those kinds of systems and three similar systems that were reviewed to improve the proposed system.

Those are Lakpura Travels(Pvt)Ltd, Sri Lanka Travel and Tourism – Travel Agency and Viator – The TripAdvisor Company.



Lakpura Travels (Pvt) Ltd is one of the most famous Sri Lankan tour operator company. They provide their service over 10+ years and also they have rave reviews and testimonials from hundreds of clients.



Sri Lanka Travel and Tourism – Travel Agency is also providing most wonderful and tour experiences and booking opportunities to the travelers.



Viator is one of the most world famous company who provides more than 200,000 bookable activities and experiences to travelers.

Above three company websites help to understand and gather more important details about tourism, traveler needs, interfaces, online tour booking experience and also they were very helpful to improve the proposed system in many ways.

2.3 Current process

There is no existing system because currently they are doing their business through Facebook page. They communicate with their customers, selecting tour packages, gathering customer details, discuss and gathering payment details all those things are doing by using WhatsApp and emails. They do not maintain proper paper works also.

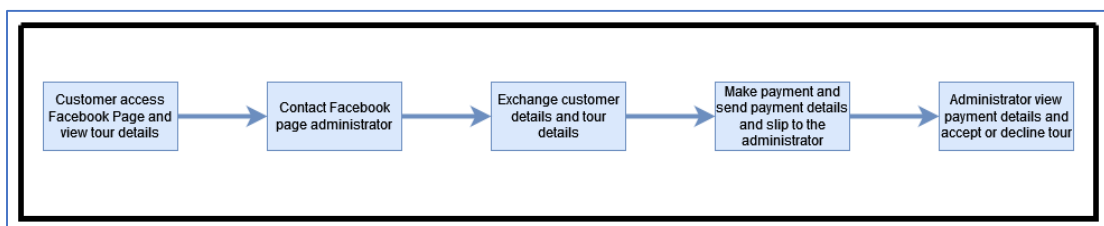


Figure 1 Current process flow

2.4 Requirement gathering techniques

Gathering client's requirements by using the fact finding techniques are the most critical part in the analysis phase. When gathering the requirements, there should be a proper way to handle these techniques. There are several fact finding techniques which can be used to collect the clear and accurate information. In this project facts were gathered by using following techniques.

1. Interviews
2. Observations
3. Sampling and documentation.

Gathering requirements using sampling and documentation, is a helpful way to get a clear idea about the system or current work process. But they do not maintain paperwork therefore it is not a good way to understand their current work process for this proposed system.

Interviews are the most commonly used technique in requirements gathering. Information was collected from the management through face to face interactions. When we carried out interviews we used both structured way and unstructured way.

Clarify and verify the facts. Feedbacks and questions were also gathered through these interviews. It becomes greater advantage to this proposed system.

2.5 Functional requirements

A functional requirement document defines the functionality of a system or one of its subsystems. It also depends upon the type of software, expected users and the type of system where the software is used.

Functional user requirements may be high-level statements of what the system should do but functional system requirements should also describe clearly about the system services in detail.

Followings are the main functional requirements of the proposed system.

Tour booking: - Allow customer to select a tour type and according to selected tour type system will display available guides, accommodations and transportations and also customer can select those as they wish. This facility available for both ready-made and customized tours. In customized tour system will allow to select tour activities as the wish.

Tracking process: - Allow customers to track their ready-made booking process and customized booking process.

Quality management: - Allow to find and update upcoming dates of interviews, vehicle check-ups and accommodation visit to the relevant users.

Helpdesk facility: - Allow to search and find right FAQs when customer needed.

Availability management: - Allow guides, accommodation owners and vehicle owners to update their availability through the system and system will display only available guides, accommodation and transportations in booking process.

Report generating: - Allow the administrator to generate reports though the system for administrator purpose. It should be accurate.

2.6 Non-Functional requirements

Nonfunctional Requirements define system attributes such as security, reliability, performance, maintainability, scalability, and usability. They serve as constraints or restrictions on the design of the system across the different backlogs. Also known as system qualities, nonfunctional requirements are just as critical as functional Epics, Capabilities, Features, and Stories. They ensure the usability and effectiveness of the entire system. Failing to meet any one of them can result in systems that fail to satisfy internal business, user, or market needs, or that do not fulfill mandatory requirements imposed by regulatory or standards agencies.

Security: - System should be achieving the security requirements. Because most of the important details- customer details, payment details are stored in this system. If that information is misplaced, then it will be a huge problem when dealing with the customers and also it will harm the company image.

Usability: - Usability requirement has been achieved by using various techniques such as easy menu navigation, searching options, attractive interfaces and use of matching colors.

Accuracy: - Accuracy is an also important nonfunctional requirement which we need to consider. If we do not consider about the accuracy, database problems can be occurred. By applying validation techniques to the user input data accuracy has been achieved in this system.

Availability: - Availability refers to a property of software that it is there and ready to carry out its task when user need it to be. Therefore, system should be always available for their users because the all business process doing through this web based system and it should be able to monitor performance live any time. System will be available 24/7 for the users.

Performance: - System should be fast and should complete the task in minimum amount of time, Data should be retrieved as soon as possible and will have a minimum response time.

2.7 Justification of the selected methodology

The waterfall process model is one of system development methodology and it was used to develop the system. It has a linear-sequential model which is simple and easy to understand and use. The waterfall model is used in small projects which have well defined requirements. The proposed system was developed by using the Waterfall model because of the clear user requirements and simplicity of use. The flow of the Waterfall model is given in figure no 2.

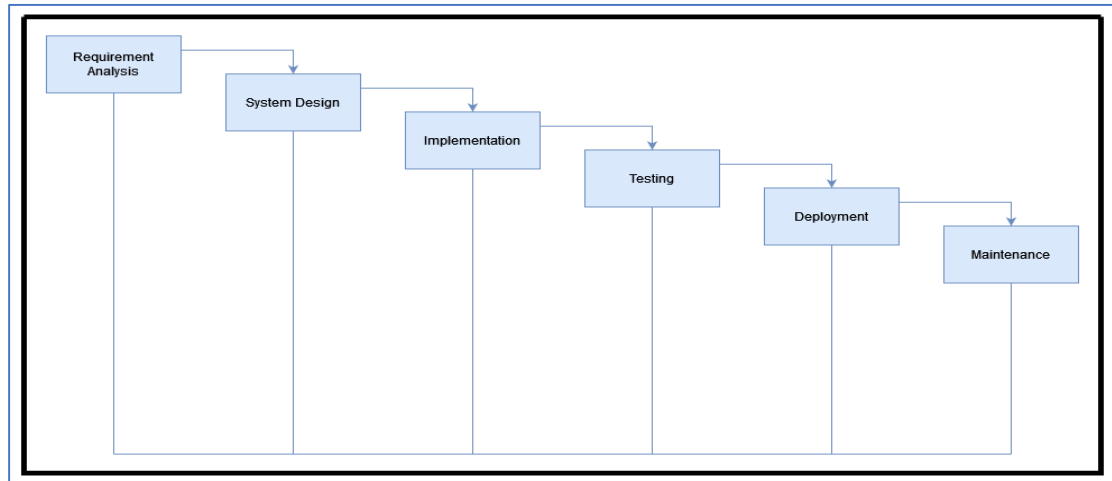


Figure 2 Waterfall model

2.8 Success factors for the system

Make sure that the system scope is well defined.

Review the user requirements simultaneously with system implementation.

Carry out user acceptance Test.

Highly concentrate on the quality of the project.

Check whether all the planned requirements are met.

Chapter 03 – Design

3.1 Introduction

An overall structure of the proposed system is discussed in the design chapter, Design methodologies, tools and techniques used in the design phase and database of the system described in this chapter along with proper use-case diagrams, Entity Relationship Diagrams and suitable UML diagrams.

3.2 Alternative solution

Alternate solution for the Tours and Travel Agency System is developing a stand-alone system, Mobile Application, or a web based system.

3.2.1 Stand-alone system

Stand-alone software is a software application that does not come bundle within another software application and does not require internet connection to run (work offline). It is not much useful for the Tours and Travel Agency System. Because their all the customers are foreign tourist. Therefore, the company cannot manage their customer without using internet. Stand-alone system installation and maintenance is harder and also costlier than to a Web Based System.

3.3 Reason for using web based solution

The company is providing their service to foreign customers. Therefore, Client specifically request for a web based solution. Web based system is easy to monitor service from any location and also the customers and other system users can access the system from anywhere anytime. Web based system maintenance is easy, because the database is centralized and everything is synchronized. The system would be platform independent and it also enables to wide range of devices to be used to access the system.

3.4 The architectural design of the system

The system is planned to develop using the Non-Object Oriented methodology as a Procedural Programming Design Concept.

A program in a procedural language is a list of instruction where each statements tells the computer to do something. It focuses on procedure (function) and algorithm is needed to perform the derived computation. When program become larger, it is divided into function and each function has clearly defined purpose. Dividing the program into functions and module is one of the cornerstones of structured programming.

There are some characteristics of procedural oriented programming

- Emphasis on doing things (Functions).
- Divides the program into multiple functions.
- Function communicate with each other by passing parameters.
- Function share global variable.
- Based upon the concept of the procedure calls.
- Follows top-down approach in program design.

3.5 Design diagrams

3.5.1 Use Case diagram for the proposed system

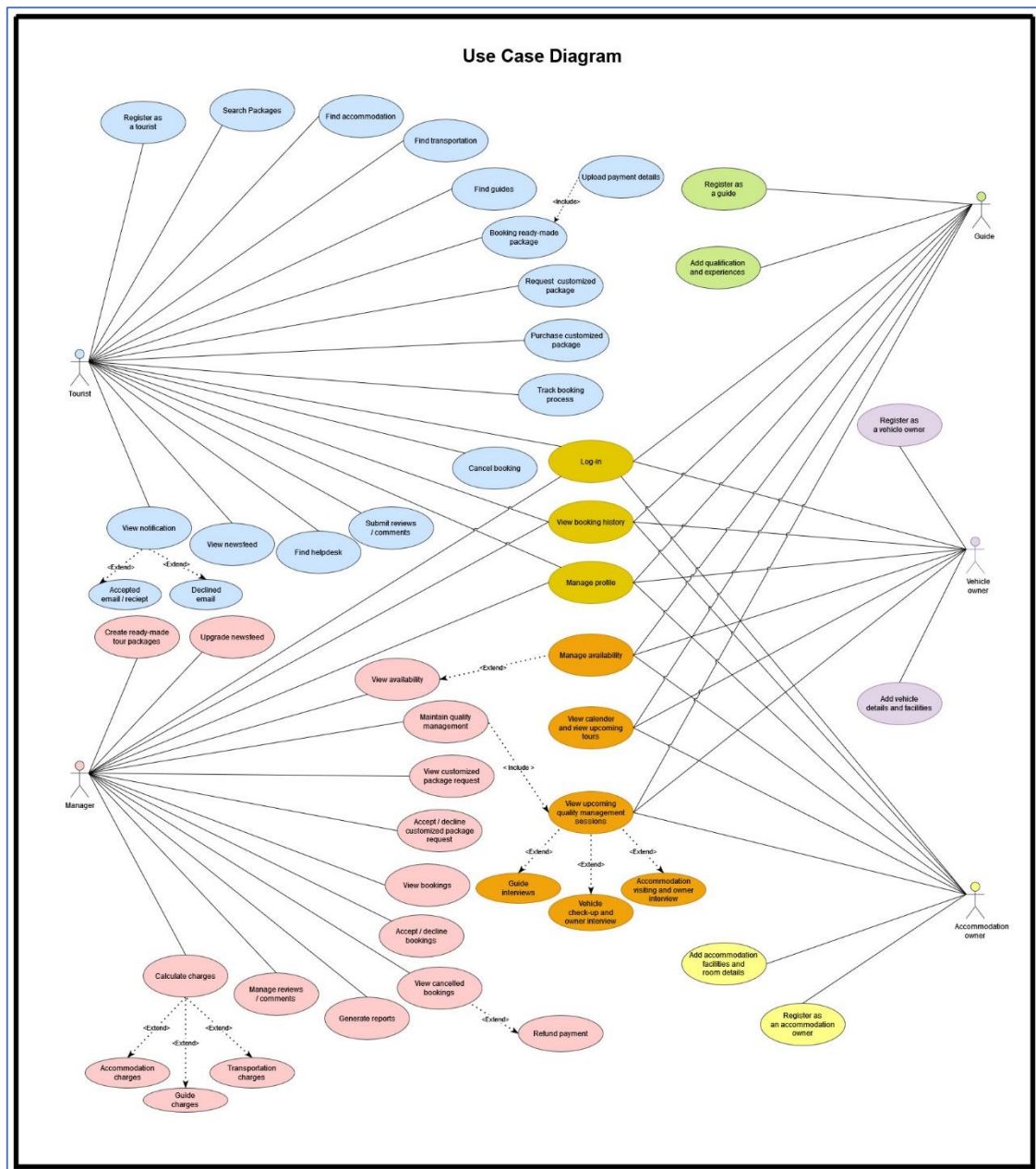


Figure 3 Use case diagram of the proposed system

3.5.2 Use Case narratives for the proposed system

Use case Number UC-01

Use Case Name	Registration	
Priority	High	
Actor	Customer (Tourist), Manager(Admin), Guide, Accommodation owner, Vehicle Owner, Staff	
Description	This use case describes how all above actors register to the Live Lanka Tours system.	
Precondition	None	
Post-condition	If the use case was successful, all above actors can register to the system.	
Basic cause of action	User Action	System Response
	1. Search website 3. Go to the registration page. 5. Fill the form according to the given details and conditions. 6. Click register button.	2. View home page. 4. View registration form. 7. Verify all the fields and accept or decline. 8. Display registration status. 9. Use case exit.
Alternate cause of action	7.1 If all fields are not filled out and not matched to the given details and conditions the system notifies the actor by appearing an error message according to the relevant field and then goes back or returns to step 5 of basic course of Action to enter again.	

3.5.3 Entity Relationship Diagram for the proposed system

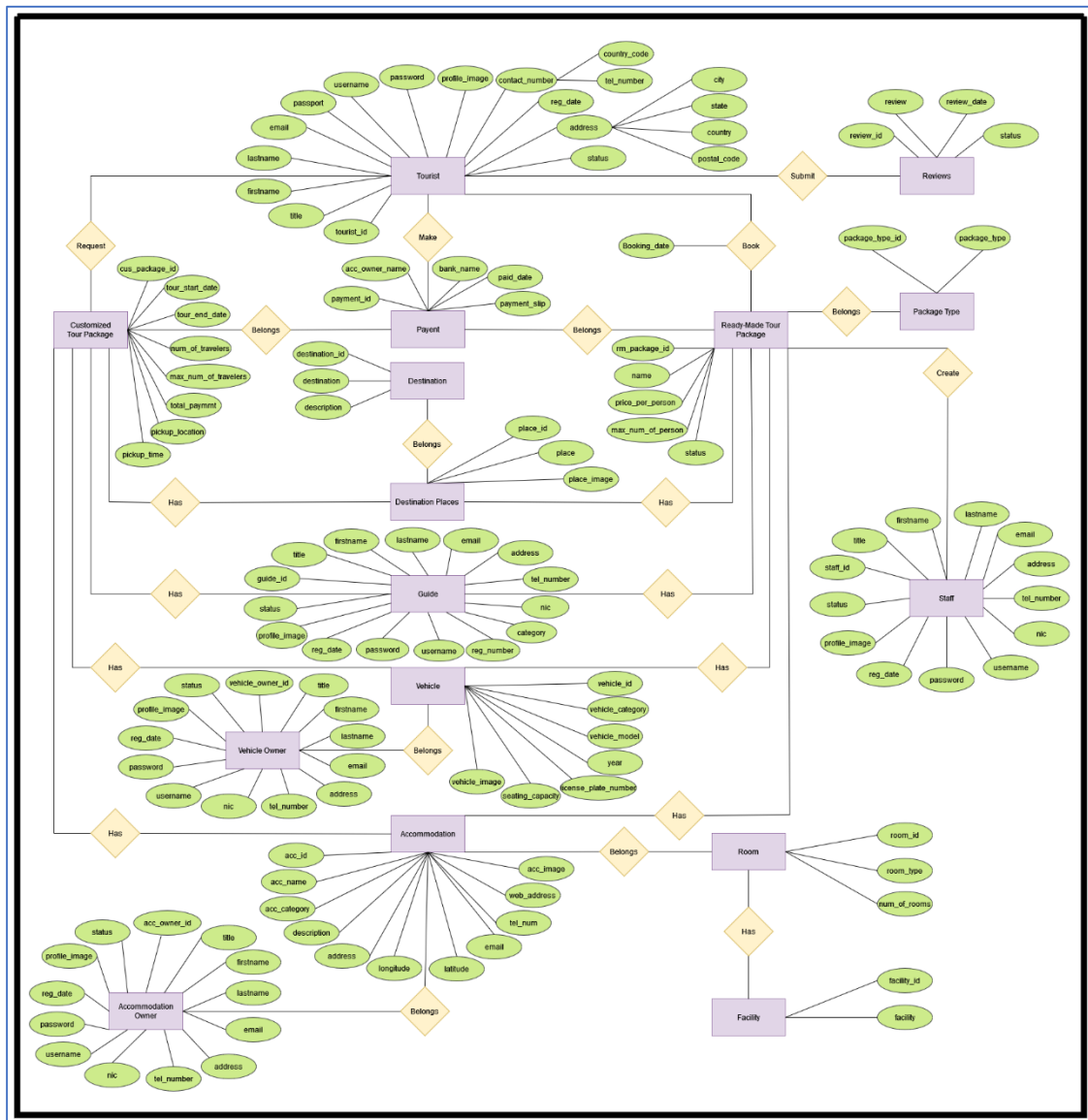


Figure 4 ER diagram of the proposed system

3.5.4 Relational Schema

The following illustrate the relational scheme in 3NF. This was done by using a technique of ERD Mapping.

Tourist (tourist_id, title, firstname, lastname, city, state, postal_code, country, country_code, tel_number, email, passport, username, password, profile_image, reg_date, status)

Guide (guide_id, title, firstname, lastname, email, address, tel_number, nic, category, registration_number, username, password, profile_image, reg_date, status)

Accommodation Owner (acc_owner_id, title, firstname, lastname, email, address, tel_number, nic, username, password, profile_image, reg_date, status)

Accommodation (acc_id, acc_name, acc_category, description, address, longitude, latitude, email, web_address, tel_number, acc_image, acc_owner_id*)

Room (room_id, num_of_rooms, acc_id*)

Facilities (facility_id, facility, room_id*)

Vehicle Owner (vehicle_owner_id, title, firstname, lastname, email, address, tel_number, nic, username, password, profile_image, reg_date, status)

Vehicle (vehicle_id, vehicle_category, vehicle_model, year, license_plate_number, seating_capacity, vehicle_image, vehicle_owner_id*)

Staff (staff_id, title, firstname, lastname, email, address, tel_number, nic, username, password, profile_image, reg_date, status)

Customized packages (cus_package_id, tour_start_date, tour_end_date, num_of_travelers, total_payment, pickup_location, pickup_time, max_num_of_person,)

Read-Made Packages (rm_package_id, name, price_per_person, max_num_of_person, status)

Package Type (package_type_id, package_type)

Destinations (destination_id, destination, description)

Destination Place (place_id, place, place_image, destination_id*)

Payment (payment_id, acc_owner_name, bank_name, paid_date, payment_slip)

Reviews (review_id, review, review_date, status)

3.5.5 Activity diagram for the proposed system

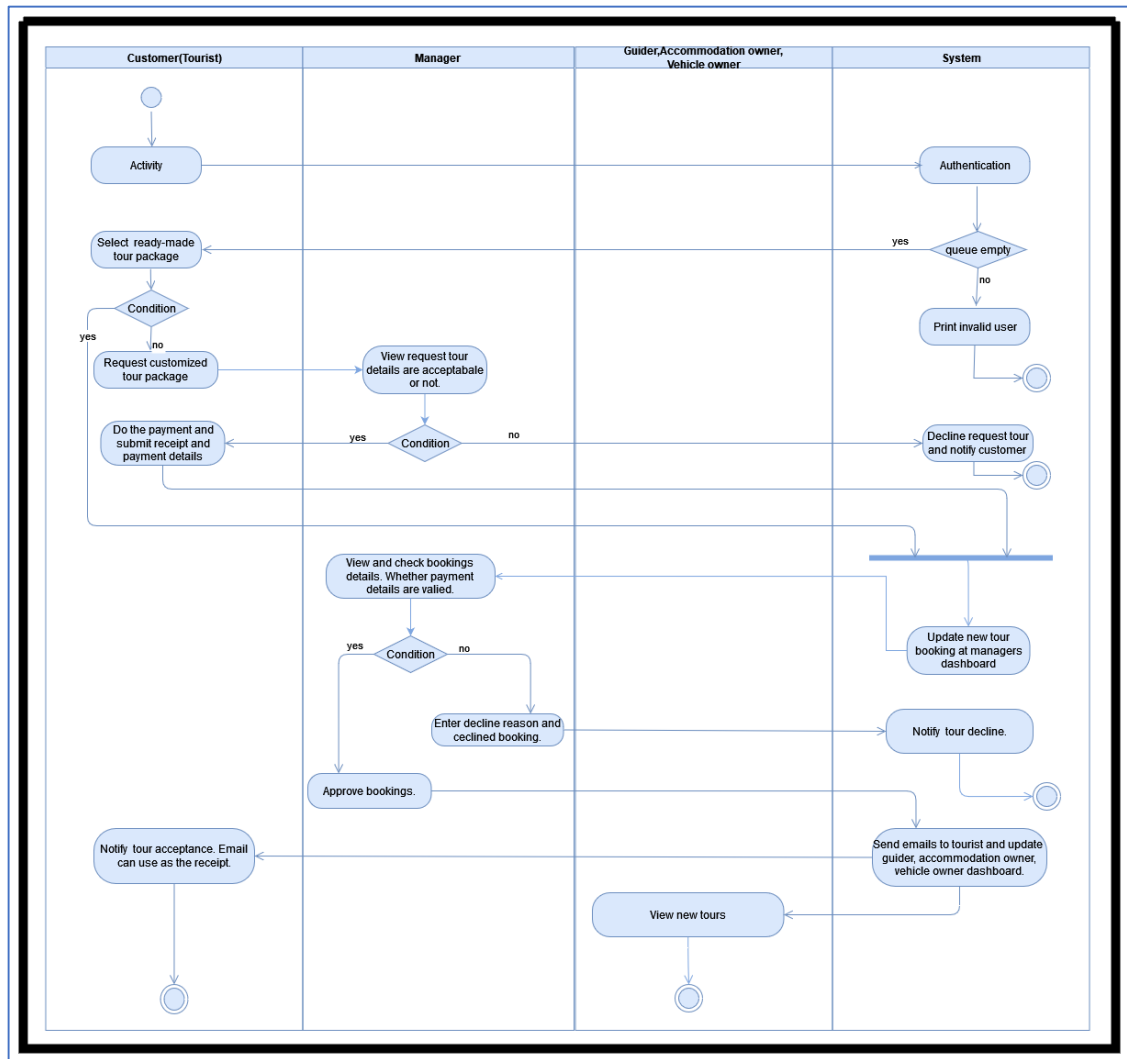


Figure 5 Activity diagram of the proposed system

3.6 User Interface Design

The important part of the system is the quality of the user interface, because of the high interactivity in interface can attract more users. Way of interacting with the system is different for each other so in order to satisfy all the customers the interface should be simply, easy to use and easy to understand instead of using complex message and heavy color. Interfaces should be made for both experts and non-experts in computer. And interfaces should be interactive as well as secure to use.

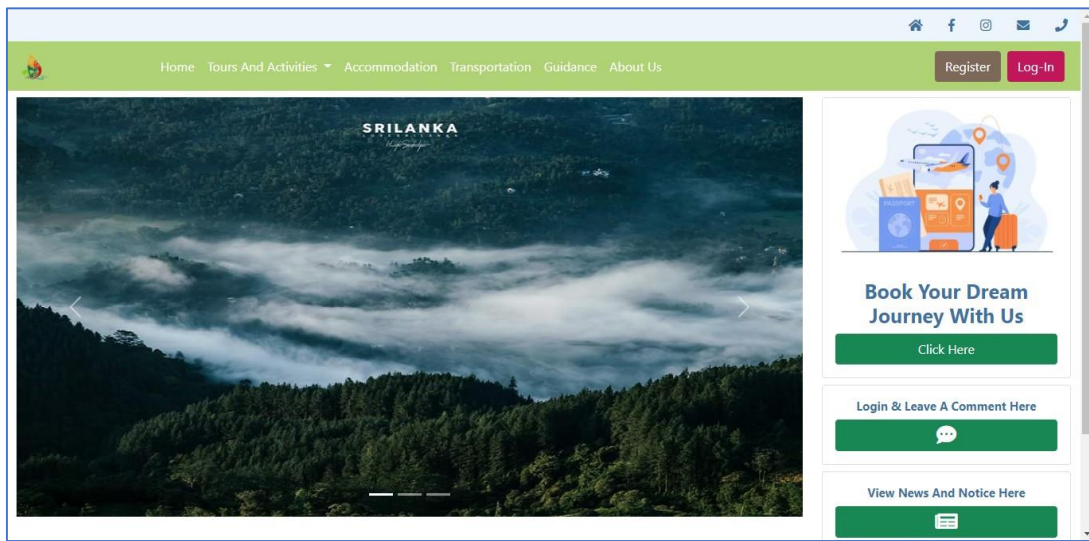


Figure 6 Home page of the proposed system

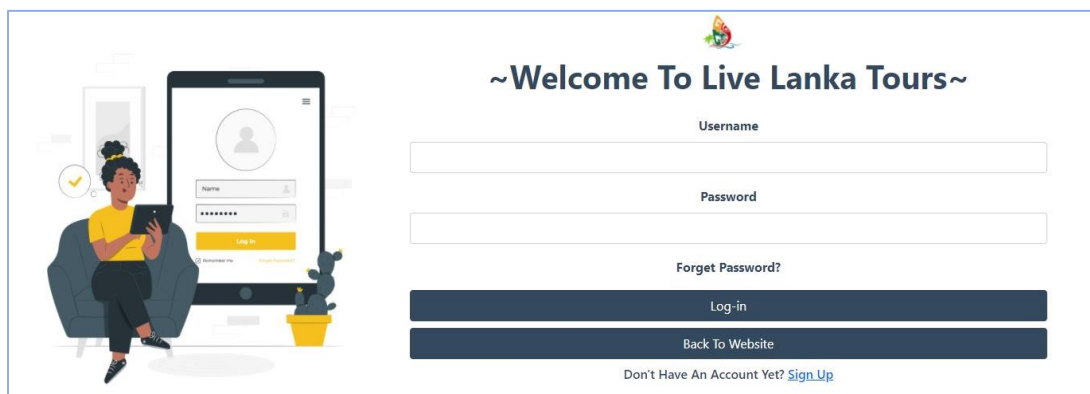


Figure 7 Log-In page of the proposed system

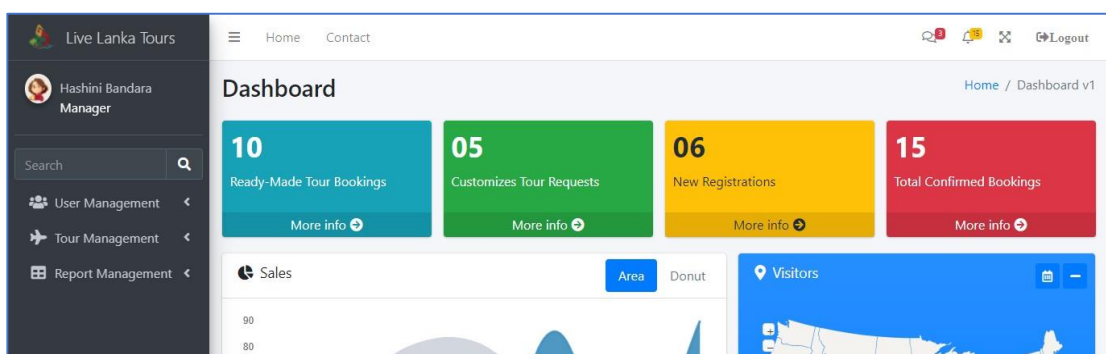


Figure 8 Manager dashboard of the proposed system

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